EVT-MARL: Risk-Averse Multi-Agent Reinforcement Learning via Extreme Value Theory

Modeling Firm Decision-Making under Heavy-Tailed Political-Economic Uncertainty

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Abstract

Standard Multi-Agent Reinforcement Learning (MARL) algorithms typically assume well-behaved, light-tailed environmental stochasticity, leaving networked multi-agent systems highly vulnerable to catastrophic cascading failures when exposed to extreme macro-level shocks. In this working paper, we introduce an **Extreme Value Theory (EVT) informed Multi-Agent Reinforcement Learning (EVT-MARL)** framework designed to model firm-level sequential decision-making under severe, heavy-tailed political-economic uncertainty, such as sanctions-driven exchange rate shocks. Building upon the single-agent Peaks-Over-Threshold Policy Gradients (POTPG) approach (Davar, Godin & Garrido, 2024), we extend EVT-based tail-risk penalization to a networked multi-agent setting. Specifically, each agent's reward function is penalized simultaneously by two Conditional Value-at-Risk (CVaR_α) terms estimated via a Generalized Pareto Distribution (GPD): (1) an *individual tail-risk penalty* capturing firm-specific exposure to fat-tailed market fluctuations, and (2) a *systemic network-wide risk penalty* capturing correlated macro-shocks that impact the entire inter-firm topology simultaneously. Grounded in empirical data comprising $5,052$ daily observations of the USD/IRR free-market exchange rate (2012–2026; excess kurtosis 17.4, GPD shape parameter $\xi \approx +0.11$), $4,260$ observations of the Tehran Stock Exchange index (TEDPIX, 2008–2026; bounded tail $\xi \approx -0.18$), and 21 discrete sanctions events (2012–2025), our multi-seed stress tests demonstrate that the EVT-aware MARL framework produces substantially lower-variance and more predictable fragility behavior under severe shocks compared to a risk-neutral baseline (achieving a fragility standard deviation of 0.07 versus 1.60 across random seeds). Furthermore, empirical crossover analysis across crisis regimes (2018–2025) identifies a critical boundary condition: the stability advantage of EVT-MARL is highly concentrated during sudden, discrete extreme shocks rather than gradual trend shifts.

Keywords: Multi-Agent Reinforcement Learning, Extreme Value Theory, Conditional Value-at-Risk, Generalized Pareto Distribution, Systemic Risk, Network Fragility, Economic Sanctions.

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1 Introduction

Sequential decision-making by economic agents operating within interconnected networks is frequently subjected to abrupt, exogenous political-economic shocks. In emerging and sanctioned economies, macroeconomic indicators—most notably free-market exchange rates—do not follow Gaussian or thin-tailed distributions. Instead, they exhibit pronounced fat-tailed behavior characterized by calm periods punctuated by sudden, severe regime shifts. When firms interact strategically within supply-chain or financial networks, such extreme shocks can trigger correlated losses that propagate through the network topology.

Standard Reinforcement Learning (RL) and Multi-Agent Reinforcement Learning (MARL) paradigms predominantly optimize expected cumulative return. While effective in stationary or mildly stochastic environments, expected-value optimization underestimates the probability and magnitude of rare, catastrophic tail events. In a networked multi-agent setting, ignoring heavy-tailed risk is particularly hazardous: an extreme decision error by a highly connected node can induce systemic contagion, destabilizing neighboring agents and degrading overall network welfare.

Recent advances in risk-sensitive single-agent RL have begun addressing rare catastrophic events by integrating statistical tools from Extreme Value Theory (EVT). Notably, NS et al. (2023) demonstrated the utility of EVT for extreme risk mitigation in single-agent control, and Davar, Godin & Garrido (2024) formulated the Peaks-Over-Threshold Policy Gradients (POTPG) algorithm to bound catastrophic tail risk via Generalized Pareto Distribution (GPD) approximations. However, single-agent formulations do not account for the topological coupling and correlated systemic losses inherent to multi-agent economic networks.

In this paper, we bridge this gap by developing the **EVT-MARL** framework, which extends EVT-informed reward penalization to a networked multi-agent environment. Our primary contributions are fourfold:

1. **Multi-Agent EVT Reward Architecture:** We formulate a risk-averse MARL reward structure that penalizes each firm agent using both an individual GPD-estimated CVaR_α term and a systemic network-wide aggregate CVaR_α term ($L_{\text{sys}} = \sum_j w_j L_j$).

2. **Empirical Calibration Across Distinct Tail Regimes:** We ground our simulation environment in over $9,300$ empirical market observations and 21 verified political-economic sanctions shocks, contrasting an unconstrained fat-tailed regime (USD/IRR free-market exchange rate, $\xi \approx +0.11$) against a regulatory price-bounded regime (TEDPIX equity index, $\xi \approx -0.18$).

3. **Topological Fragility Quantification:** We evaluate network robustness using a second-order mathematical fragility index ($\partial^2 L_i / \partial \theta^2$), demonstrating via multi-seed stress testing that EVT-MARL compresses cross-seed fragility standard deviation from 1.60 (in risk-neutral baselines) down to 0.07 under severe shocks.

4. **Boundary Condition Identification and Open-Source Benchmark:** Through empirical crossover analysis on 2018–2025 crisis episodes, we show that EVT-informed penalization provides its primary advantage during sudden, discrete extreme events rather than gradual trends, and we provide a complete open-source simulation suite with an interactive NetworkX and Streamlit dashboard.

2 Related Work

2.1 Risk-Sensitive and Distributional Reinforcement Learning

Traditional risk-averse RL incorporates risk measures such as variance, exponential utility, or Conditional Value-at-Risk (CVaR) into the optimization objective. Empirical CVaR estimation in standard RL typically relies on sample quantiles from observed trajectory returns. However, when the underlying loss distribution is heavy-tailed, empirical quantiles suffer from severe sample inefficiency and high estimation variance in the far tail, precisely where catastrophic failures occur.

2.2 Extreme Value Theory in Reinforcement Learning

Extreme Value Theory provides a rigorous asymptotic foundation for modeling the stochastic behavior of extreme deviations from the median of probability distributions. By applying the Pickands–Balkema–de Haan theorem, exceedances above a high threshold can be modeled parametrically using the Generalized Pareto Distribution (GPD), enabling reliable extrapolation into the tail beyond observed historical maxima.

NS et al. (2023) introduced EVT into reinforcement learning to mitigate extreme tail risks in single-agent settings. Subsequently, Davar, Godin & Garrido (2024) proposed the Peaks-Over-Threshold Policy Gradients (POTPG) framework, integrating GPD-based tail-risk estimates directly into policy gradient updates to avoid catastrophic states. Our work builds directly upon the POTPG paradigm, extending it from an isolated single-agent setting to an interactive, graph-structured multi-agent system where agents face both idiosyncratic tail shocks and correlated systemic contagion.

3 Empirical Data and Tail-Risk Characterization

To ensure that our multi-agent simulation reflects authentic heavy-tailed political-economic uncertainty rather than purely synthetic noise, the environment is calibrated using three empirical datasets spanning up to 18 years of market and geopolitical history. Table 1 summarizes the empirical datasets and their estimated tail characteristics.

Table 1: Summary of Empirical Datasets and Statistical Tail Behavior

Dataset	**Coverage**	**Description**	**Empirical Tail Behavior**
USD/IRR Free-Market Rate	2012–2026 ($5,052$ obs.)	Real exchange rate shocks	Fat-tailed (Excess kurtosis: 17.4, GPD $\xi \approx +0.11$)
Sanctions Events	2012–2025 (21 events)	OFAC, Iran Primer, EU Consilium	Discrete extreme political-economic shocks
TEDPIX (Tehran Stock Exchange)	2008–2026 ($4,260$ obs.)	Market response index	Bounded tail ($\xi \approx -0.18$) due to regulatory daily price bands

3.1 Contrast Between Fat-Tailed and Bounded Regimes

A central empirical insight emerging from the data analysis ('01_data_analysis.ipynb') is the structural divergence in tail behavior between unconstrained currency markets and regulated equity markets under identical geopolitical shocks:

- **Unconstrained Fat-Tailed Regime (USD/IRR):** Over $5,052$ trading days between 2012 and 2026, the free-market exchange rate exhibits extreme leptokurtic behavior with an excess kurtosis of 17.4. Fitting a Generalized Pareto Distribution via the Peaks-Over-Threshold (POT) method yields a positive shape parameter of $\xi \approx +0.11$. A positive shape parameter ($\xi > 0$) confirms a heavy-tailed Fréchet-type domain of attraction, where extreme losses decay as a power law and moments above $1/\xi$ become infinite.
- **Regulated Bounded-Tail Regime (TEDPIX):** Across $4,260$ observations from 2008 to 2026, the Tehran Stock Exchange index exhibits a negative GPD shape parameter of $\xi \approx -0.18$ (Weibull-type bounded tail). This truncation is directly attributable to regulatory interventions, specifically mandatory daily price fluctuation bands that artificially cap single-day equity losses and prevent immediate tail realizations, shifting adjustment pressures toward liquidity freezes or multi-day trends.
- **Discrete Political Shocks (Sanctions Dataset):** We compile 21 major geopolitical and economic sanctions events between 2012 and 2025 from official primary records (including U.S. Department of the Treasury OFAC designations, the Iran Primer timeline, and European Union Consilium decisions). These events serve as discrete exogenous shock triggers during stress-testing replay episodes.

4 Methodological Framework

4.1 Peaks-Over-Threshold GPD Tail Estimation

Let L denote a random loss variable experienced by an agent. Given a high threshold u, the distribution of exceedances $Y = L - u$ conditional on $L > u$ converges asymptotically to the Generalized Pareto Distribution with cumulative distribution function:

$$G_{\xi,\sigma}(y) = 1 - \left(1 + \frac{\xi y}{\sigma}\right)^{-1/\xi}, \quad \text{for } \xi \neq 0, \tag{1}$$

where $\sigma > 0$ is the scale parameter and $\xi \in \mathbb{R}$ is the shape parameter governing tail heaviness. Using the fitted GPD parameters $(\hat{\xi}, \hat{\sigma})$ and the empirical exceedance probability above u, the Conditional Value-at-Risk at confidence level $\alpha \in (0,1)$, denoted $\text{CVaR}_\alpha(L)$, is computed analytically in closed form ('src/evt_utils.py'), providing a parametric tail-risk estimate that remains stable even at extreme quantiles.

4.2 Dual CVaR-Penalized Multi-Agent Reward Function

We consider a networked multi-agent system represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where each node $i \in \mathcal{V} = \{1, \ldots, N\}$ corresponds to a decision-making firm ('FirmAgent') and edges $\mathcal{E}$ capture inter-firm economic linkages. At each discrete time step, firm i observes state s, selects action a_i (while other agents select joint action a_{-i}), and receives a raw operational reward $R(s, a_i)$ alongside a stochastic loss realization L_i.

Whereas standard MARL maximizes expected cumulative reward, our EVT-MARL framework modifies each agent's immediate objective by penalizing both idiosyncratic and systemic extreme tail losses:

$$R_{\text{MARL}}(s, a_i, a_{-i}) = R(s, a_i) - \lambda_1 \cdot \text{CVaR}_\alpha(L_i) - \lambda_2 \cdot \text{CVaR}_\alpha(L_{\text{sys}}), \tag{2}$$

where:

- L_i denotes the individual loss distribution of agent i, with $\text{CVaR}_\alpha(L_i)$ estimated via GPD fitting on historical exchange rate shocks.
- $L_{\text{sys}} = \sum_{j=1}^{N} w_j L_j$ represents the network-wide aggregate loss weighted by topological importance weights w_j, capturing the correlated nature of macro-shocks that strike the entire network simultaneously.
- $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$ are hyperparameter weights balancing expected operational return against individual tail-risk aversion and systemic network stability, respectively.

4.3 Second-Order Topological Fragility Index

To rigorously evaluate the structural resilience of each agent and the overarching multi-agent topology under varying shock magnitudes θ, we define a second-order fragility index ('src/fragility.py') inspired by convexity analysis of loss functions:

$$\text{Fragility}_i = \frac{\partial^2 L_i}{\partial \theta^2}. \tag{3}$$

This second partial derivative measures the sensitivity and acceleration of agent i's loss with respect to changes in macro-shock intensity θ:

- **Positive ($\text{Fragility}_i > 0 \rightarrow$ Fragile):** Convex loss response; losses accelerate non-linearly as shock magnitude θ increases, indicating vulnerability to cascading failure.
- **Near Zero ($\text{Fragility}_i \approx 0 \rightarrow$ Robust):** Linear or invariant loss response; the agent absorbs extreme shocks predictably without non-linear amplification.
- **Negative ($\text{Fragility}_i < 0 \rightarrow$ Antifragile):** Concave loss response; marginal loss decelerates or the agent adapts advantageously under heightened volatility.

5 Empirical Evaluation and Multi-Seed Stress Testing

We evaluate the EVT-MARL architecture against a standard risk-neutral MARL baseline ($\lambda_1 = 0, \lambda_2 = 0$) across multiple random seeds using tabular Q-learning agents interacting on a NetworkX graph topology. The experimental protocol progresses from baseline training ('02_baseline_marl.ipynb') and EVT-penalized training ('03_risk_aware_marl.ipynb') to historical crisis scenario replay and multi-seed stress testing ('04_stress_testing.ipynb').

5.1 Variance Compression and Fragility Stability

Table 2 presents the core comparative finding across multi-seed simulation runs under severe shock scenarios.

As shown in Table 2, risk-neutral MARL agents exhibit severe instability across random seeds when subjected to heavy-tailed shocks, yielding a fragility standard deviation of 1.60. Because standard Q-learning agents optimize average returns during predominantly

Table 2: Comparison of Fragility Behavior Across Multi-Seed Stress Tests

Model Architecture	Reward Specification	Fragility SD	Observed Network Behavior under Severe Shocks
Risk-Neutral Baseline	$\lambda_1 = 0,\ \lambda_2 = 0$	1.60	High cross-seed variance; prone to erratic fragility spikes and cascading topological instability.
EVT-Aware MARL	$\lambda_1 > 0,\ \lambda_2 > 0$ (GPD-CVaR)	**0.07**	Substantially lower variance (95.6% reduction in SD); highly predictable, robust fragility trajectories across seeds.

calm training steps, a single extreme draw from the fat tail ($\xi \approx +0.11$) destabilizes their policy updates and propagates through the network. Conversely, incorporating individual and systemic GPD-based CVaR$_\alpha$ penalties in EVT-MARL compresses the cross-seed fragility standard deviation to 0.07, ensuring consistent, bounded fragility behavior near the robust regime (Fragility$_i \approx 0$).

5.2 Crossover Analysis and Boundary Conditions (2018–2025)

To understand precisely when EVT-based penalization outperforms standard MARL, we conduct an empirical crossover analysis replaying historical sanctions episodes between 2018 and 2025. This analysis reveals an important operational boundary condition:

- **Sudden, Discrete Extreme Events:** During abrupt political-economic shocks (such as sudden OFAC announcements or acute currency devaluations), EVT-MARL demonstrates a decisive stability advantage over the risk-neutral baseline, preventing nonlinear loss acceleration.
- **Gradual Trends:** During prolonged, low-volatility drift or bounded-tail regimes (analogous to the regulated TEDPIX environment), the performance gap narrows, as standard temporal-difference learning has sufficient time to adapt incrementally without encountering tail discontinuities.

6 Software Architecture and Interactive Simulation

To support full reproducibility and interactive inspection of network dynamics, the complete framework is implemented in Python ($3.8+$) and organized into a modular open-source repository ('EVT-MARL-Sanctions'):

- **Agent Module (`src/agents.py`):** Implements the `FirmAgent` class utilizing tabular Q-learning with customizable risk-aversion coefficients (λ_1, λ_2).
- **Network Environment (`src/environment.py`):** Manages the multi-agent simulation environment, inter-firm topological coupling via `NetworkX`, and shock injection dynamics.

- **EVT and Fragility Utilities (`src/evt_utils.py`, `src/fragility.py`):** Provides POT threshold selection, GPD maximum-likelihood estimation, closed-form CVaR_α calculation, and finite-difference computation of the second-order fragility index $\partial^2 L_i / \partial \theta^2$.
- **Interactive Dashboard (`app/main.py`):** A real-time `Streamlit` web application that visualizes the NetworkX firm topology colored by node fragility, replays historical sanctions shocks, and compares EVT-MARL against risk-neutral baselines side by side.

7 Conclusion and Future Work

This working paper introduced EVT-MARL, a risk-averse multi-agent reinforcement learning framework that integrates Extreme Value Theory into networked sequential decision-making. By penalizing agent rewards with both individual and systemic GPD-derived CVaR_α terms, our approach mathematically bounds sensitivity to heavy-tailed political-economic shocks and reduces cross-seed fragility standard deviation from 1.60 to 0.07.

Future research will extend this foundational framework in two directions: (1) scaling from tabular Q-learning to deep multi-agent reinforcement learning via a risk-sensitive **Multi-Agent Proximal Policy Optimization (MAPPO)** architecture capable of handling continuous, high-dimensional state-action spaces, and (2) deriving formal theoretical sample-complexity and convergence bounds for networked GPD policy gradients under non-stationary regime shifts.

References

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